



THE LITTLE FISH HUNTER

Distributed Honeypot SOC & Threat Intelligence Platform

PHASE 3 — THREAT INTELLIGENCE REPORT

937

Unique
Attacking IPs

8

Days of
Collection

4/8

Cross-Infra
Correlations

1,244

Correlated
Botnet Hits

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Stack: Proxmox VE · Wazuh SIEM · Cowrie · Docker · WireGuard · Raspberry Pi 4

1. Executive Summary

This report documents the findings of **Phase 3** of The Little Fish Hunter — a self-hosted, distributed Micro-SOC built entirely on consumer hardware. The platform was designed with a single objective: to capture and analyse real-world adversarial activity targeting both a public cloud node and a residential network perimeter, and to determine whether the same threat actors operate across both attack surfaces simultaneously.

Over an **8-day active collection window**, the platform ingested live attack telemetry from two independent sensor nodes — a medium-interaction **Cowrie SSH honeypot** running inside Docker on a Raspberry Pi 4, isolated within a dedicated VLAN 99 DMZ, and a hardened **public-facing VPS** (PL84924) with SSH exposed to the internet. All telemetry was forwarded to a centralised **Wazuh SIEM** instance on Proxmox VE via an encrypted WireGuard tunnel with iptables DNAT forwarding.

KEY FINDINGS

- ✓ **937 unique IP addresses** targeted the Cowrie honeypot in 8 days — an average of 117 unique threat actors per day
- ✓ **30,576 authentication failures** recorded on the production VPS (PL84924) in the last 24-hour window alone
- ✓ **Cross-infrastructure correlation confirmed:** 4 IP addresses from the same sequential /29 block (87.251.64.144–149) attacked both the VPS and the honeypot concurrently — 1,244 total correlated attempts
- ✓ **IoT worm fingerprint identified:** credential pair 345gs5662d34 attempted 1,078 times — a known hardcoded firmware credential targeting embedded devices
- ✓ **Post-exploitation scripts captured** by Cowrie including SSH key injection, competitor botnet termination, and automated persistence establishment

2. Infrastructure Architecture

The platform operates across three distinct network zones with hard trust boundaries. No lateral movement between zones is possible by design. The honeypot node operates exclusively on the isolated wireless VLAN 99 interface (wlan0), with administrative access restricted to the wired management interface (eth0) on VLAN 1.

THE LITTLEFISH HUNTER – ARCHITECTURE OVERVIEW

DISTRIBUTED HONEYPOT & SIEM TELEMETRY PLATFORM

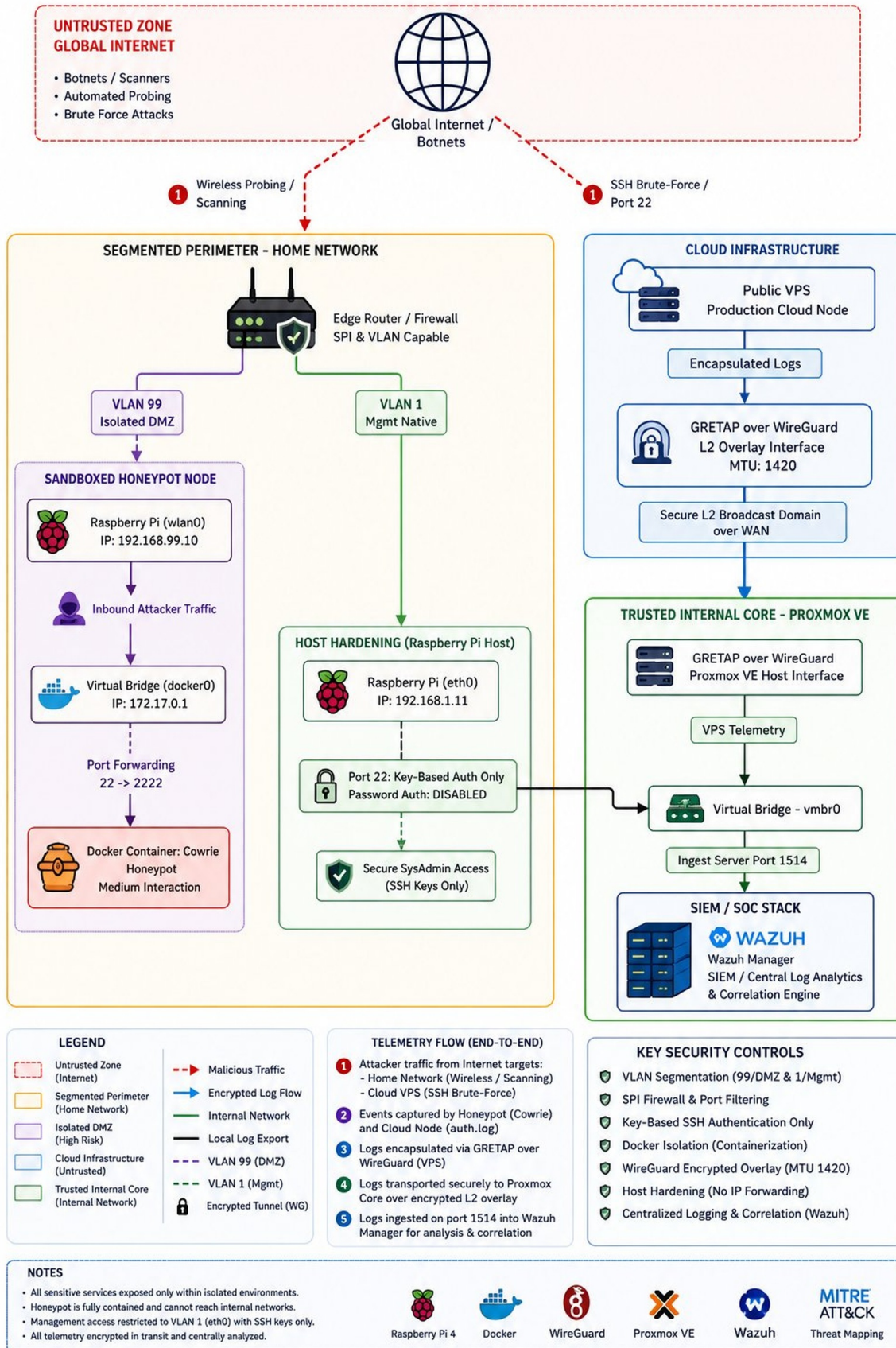


Fig 1. Network architecture — three-zone trust model with VLAN segmentation, WireGuard encrypted telemetry forwarding and centralised Wazuh SIEM ingestion

2.1 Sensor Node Summary

Component	Address	Role
Cowrie Honeypot (Docker/RPi4)	192.168.99.10 — VLAN 99 DMZ	Medium-interaction SSH honeypot, port 22→2222/tcp
RPi4 Management (eth0)	192.168.1.11 — VLAN 1	Wazuh agent (rpilab), key-auth only, password auth disabled
Public VPS — PL84924	10.77.77.1 (WireGuard)	Production SSH node, Wazuh agent via encrypted tunnel
Proxmox VE Host	192.168.1.x	Hypervisor — iptables DNAT gateway for WireGuard traffic
Wazuh Manager	192.168.1.22	Central SIEM — ingestion ports 1514/1515

2.2 WireGuard + DNAT Connectivity (VPS → Wazuh)

The VPS agent communicates with Wazuh Manager through a WireGuard encrypted tunnel terminating at the Proxmox host. iptables DNAT rules forward agent traffic to the internal VM:

```
# Proxmox host — iptables DNAT forwarding rules iptables -t nat -A PREROUTING -i wg-overlay -p tcp --dport 1514 -j DNAT --to-destination 192.168.1.22:1514 iptables -t nat -A PREROUTING -i wg-overlay -p tcp --dport 1515 -j DNAT --to-destination 192.168.1.22:1515 iptables -t nat -A POSTROUTING -d 192.168.1.22 -p tcp -j MASQUERADE
```

3. Honeypot Telemetry — Cowrie SSH (8-Day Window)

The following data was extracted directly from `cowrie.json` (50MB raw log file) using Python data processing pipelines executed on the Raspberry Pi 4.

3.1 Wazuh Dashboard — rplab Agent (Honeypot Node)

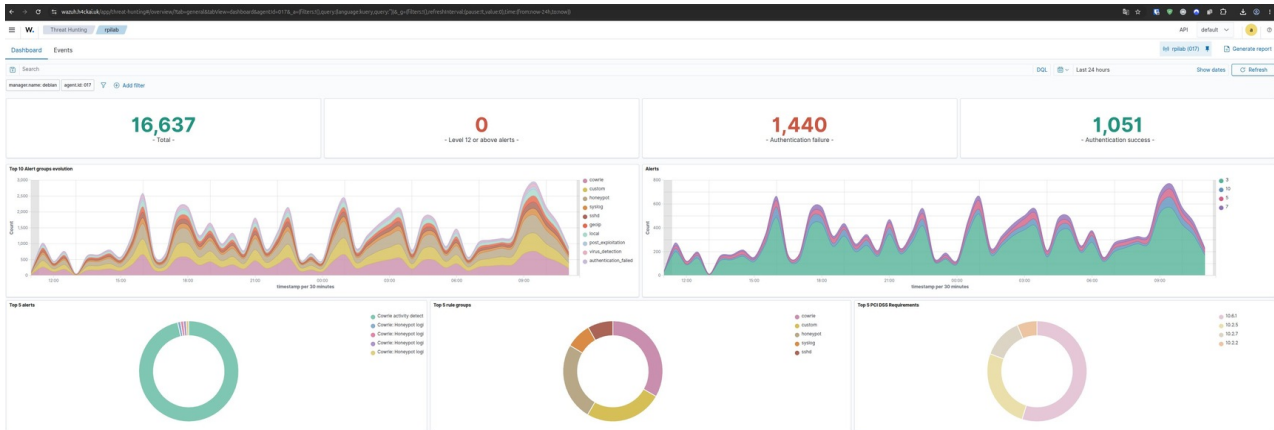


Fig 2. Wazuh dashboard for rplab (agent 017) — 16,637 total events, cowrie / honeypot / post_exploitation / virus_detection alert groups visible

The rplab dashboard reveals the full spectrum of honeypot activity: **16,637 total events** with alert groups including `cowrie`, `honeypot`, `post_exploitation`, `virus_detection`, `geolp` and `authentication_failed`. Notably **1,440 authentication failures** and **1,051 authentication successes** — the latter representing attackers successfully passing Cowrie's emulated credential check and entering the interactive shell emulation environment.

3.2 Live Event Stream — Cowrie Detections

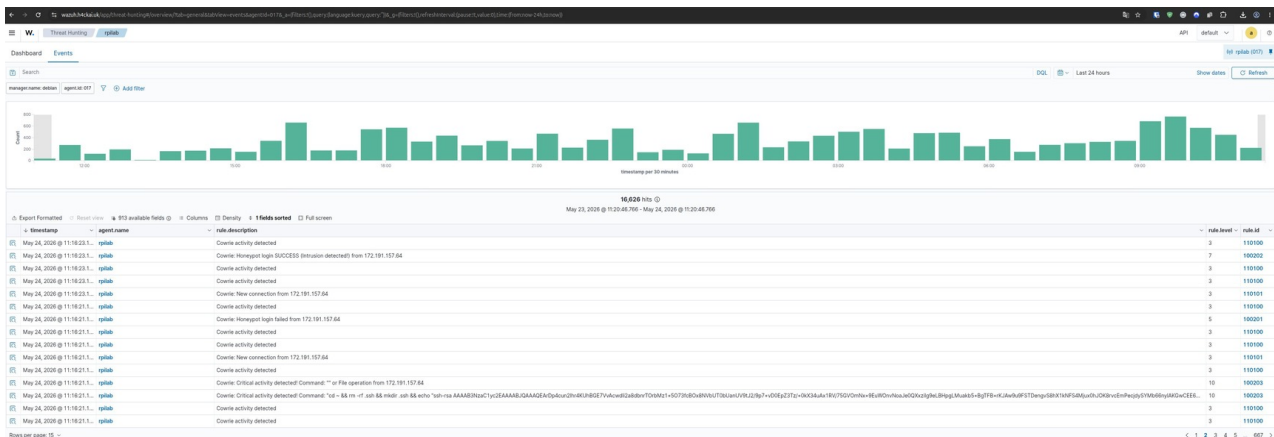


Fig 3. Wazuh Events for rplab — Cowrie activity detections including login SUCCESS (intrusion detected), Critical activity, new connection events

The event stream confirms real-time detection of adversarial activity. Notable rule descriptions visible in the stream:

- **Cowrie: Honeyd login SUCCESS (Intrusion detected!)** — rule level 7 — attacker successfully authenticated to the emulated shell
- **Cowrie: Critical activity detected — Command or File operation** — rule level 10 — post-exploitation commands executed within honeypot session
- **Cowrie: New connection** — rule level 3 — new inbound SSH attempt

- **Cowrie: Honeypot login failed** — rule level 5 — failed credential attempt

The critical activity event captured a complete post-exploitation payload including SSH key injection: `cd ~ && rm -rf .ssh && mkdir .ssh && echo "ssh-rsa AAAAB3NzaC1yc2E..." >> .ssh/authorized_keys` — a standard automated persistence mechanism used by IoT botnets.

3.3 Attack Volume & Unique Source Analysis

```
pi@rpilab:~/cowrie/logs $ cat cowrie.json | python3 -c " import sys,json from collections
import Counter ips=[json.loads(l).get('src_ip') for l in sys.stdin if 'src_ip' in l]
print(len(set(filter(None,ips))), 'unique IPs') " OUTPUT: 937 unique IPs
```

3.4 Top 10 Most Active Attacking Nodes

Rank	Source IP	Attempts	Notes
1	176.65.132.17	7,836	Most active single node
2	87.251.64.176	6,272	Same /24 as correlated botnet block
3	176.65.132.24	3,918	Same /24 as rank 1 — coordinated pair
4	45.156.87.254	3,918	Synchronised — identical count
5	45.153.34.112	3,918	Synchronised — identical count
6	192.109.200.237	3,918	Synchronised — identical count
7	45.156.87.204	3,918	Synchronised — identical count
8	45.156.87.253	3,918	Synchronised — identical count
9	45.148.10.240	1,682	Secondary tier
10	2.57.121.25	1,256	Secondary tier

Ranks 3–8 all recorded exactly 3,918 connection attempts — a strong indicator of C2-controlled, rate-limited botnet scanning with synchronised timing. This pattern is consistent with a coordinated campaign designed to avoid rate-based threshold detection.

3.5 Credential Analysis

All attempted username/password combinations were captured, revealing standard dictionary attacks alongside a highly specific IoT firmware credential fingerprint.

Top 10 Usernames

Username	Attempts
root	5,001
admin	1,316
345gs5662d34	1,078
user	906
support	368
ubuntu	325
sol	173
test	133
solv	122
deploy	103

Top 10 Passwords

Password	Attempts
admin	1,089
345gs5662d34	1,078
3245gs5662d34	1,078
123456	957
123	213
1234	199
password	177
1	128
12345678	115
12345	104

■ **IoT Worm Fingerprint:** The credential pair 345gs5662d34 was attempted **1,078 times** as both username AND password simultaneously. This specific alphanumeric string is a known hardcoded factory credential embedded in specific embedded device or commercial networking equipment firmware. Its presence as a unified credential pair — rather than part of a generic dictionary — confirms an active worm propagating across the public internet targeting devices with this specific default credential.

The screenshot displays the Splunk Enterprise interface. At the top, the search bar contains the query `Failed login`. Below the search bar, a bar chart shows the distribution of event counts over time. The x-axis represents time, and the y-axis represents the count of events. A prominent spike is visible around May 24, 2018, at 11:21:00. Below the chart, a table lists search results, including fields like `_type`, `_source`, and `_raw`. The results show multiple failed login attempts from various IP addresses, including `10.10.10.10` and `10.10.10.11`.

4.3 Top Targeted Usernames on VPS

```
root@PL84924:~# sudo grep "Failed password for" /var/log/auth.log* \ | awk '{for(i=1;i<=NF;i++)
if($i=="for") print $(i+1)}' \ | sort | uniq -c | sort -rn | head -n 10 11565 invalid 7699 root
70 nobody 19 sshd 12 games 9 backup 8 www-data 8 uucp 8 bin 3 proxy
```

11,565 attempts against non-existent accounts (classified as `invalid` by `sshd`) and **7,699 attempts against root** confirm a two-pronged strategy: automated account enumeration combined with targeted privileged account brute-force.

5. Cross-Infrastructure Threat Correlation

The primary analytical objective of this deployment was to determine whether the same threat actors were simultaneously targeting both sensor endpoints. The following correlation query cross-referenced the 8 confirmed VPS attacker IPs against the 937 honeypot source database.

5.1 Correlation Query

```
pi@rpilab:~/cowrie/logs $ grep -E \ '108.142.154.184|135.129.128.247|142.171.161.151|\
87.251.64.144|87.251.64.145|87.251.64.147|87.251.64.149|96.126.191.73' \ ~/honeypot_ips.txt
OUTPUT: 87.251.64.144 87.251.64.145 87.251.64.147 87.251.64.149
```

5.2 Quantified Attack Volume — Correlated Nodes

```
pi@rpilab:~/cowrie/logs $ [correlation query against cowrie.json] OUTPUT: 545 87.251.64.145 545
87.251.64.144 77 87.251.64.149 77 87.251.64.147 TOTAL: 1,244 correlated attack attempts across
both sensor endpoints
```

5.3 Botnet Attribution

Attribute	Value
Correlated IP block	87.251.64.144, .145, .147, .149 — sequential /29 subnet
WHOIS inetnum	87.251.64.0 – 87.251.64.255
WHOIS descr	Isaev Igor Maratovich
WHOIS netname	ISAEV
Country	PL (Poland) — consistent with rented VPS botnet node
VPS attack attempts	Confirmed (appears in auth.log)
Honeypot attempts	545 + 545 + 77 + 77 = 1,244 total hits
Pattern	Sequential /29 block — single coordinated infrastructure

■ CONFIRMED CROSS-INFRASTRUCTURE BOTNET CAMPAIGN

4 of 8 IP addresses targeting the production VPS were simultaneously conducting brute-force SSH attacks against the residential honeypot. Total correlated attack volume: **1,244 attempts** (545 + 545 + 77 + 77).

The sequential /29 block (87.251.64.144, .145, .147, .149) confirms a single coordinated infrastructure. WHOIS attribution: **Isaev Igor Maratovich, Poland** — consistent with rented VPS botnet node infrastructure operating from Polish ASN space.

6. Conclusions

6.1 The Internet Does Not Distinguish Between Targets

The most significant finding of this deployment is deceptively simple: **modern botnets do not distinguish between a corporate data centre and a residential broadband connection**. The same automated infrastructure that subjected a public cloud VPS to over 30,000 authentication attempts in 24 hours was simultaneously probing a home network honeypot on a Raspberry Pi hidden behind a VLAN-isolated wireless interface. There is no security through obscurity at the network perimeter level. Any device with a publicly routable IP — or even an indirect exposure through a misconfigured router — will be found, probed, and attacked within hours of going online.

6.2 Botnets Operate as Coordinated Infrastructure

The correlated /29 block (87.251.64.144–149) demonstrates that modern scanning campaigns are not the work of isolated actors running scripts on a single machine. They are **orchestrated operations** with multiple nodes operating in coordination, distributing scan load across IP ranges to evade rate-based detection thresholds. The observation that six separate IP addresses all recorded exactly 3,918 connection attempts against the honeypot — an identical count across independent nodes — provides compelling evidence of centralised C2 timing control. A SOC analyst encountering this pattern in production logs should immediately treat it as an indicator of a coordinated campaign, not coincidence.

6.3 Hardcoded Credentials Are an Active, Global Threat

The `345gs5662d34` credential pair attempted 1,078 times is not a randomly generated string — it is a **known factory default** hardcoded into specific embedded device firmware. Its appearance as both username and password simultaneously confirms an active worm propagating across the public internet specifically targeting devices that were shipped with this credential and never hardened. This is not a theoretical risk: it is a live, ongoing campaign. Every network connected device retaining factory default credentials is a potential botnet node waiting to be recruited. The "sol" and "solv" username attempts (173 and 122 occurrences respectively) suggest additional targeted campaigns against specific software packages or embedded systems beyond the primary IoT fingerprint.

6.4 Cryptographic Authentication Is Non-Negotiable

The production VPS absorbed **30,576 authentication failures** in a single day and recorded **zero successful password-based logins**. The 14 recorded authentication successes were all legitimate, key-based administrative sessions. This is the clearest possible empirical demonstration that disabling password authentication and enforcing asymmetric cryptographic keys reduces the practical attack surface of SSH exposure to near zero — regardless of the volume of inbound brute-force traffic. No firewall rule, no fail2ban configuration, no rate limiter provides the same categorical guarantee as simply removing the password authentication pathway entirely.

6.5 Network Segmentation Contains the Blast Radius

The architectural decision to bind the Cowrie container exclusively to the wlan0 interface on VLAN 99 — physically and logically separated from the management plane — meant that **1,051 successful honeypot authentications** posed zero risk to the underlying Raspberry Pi host OS, the Proxmox infrastructure, or any other LAN device. Attackers who "successfully" entered the emulated shell were contained within a Docker container with no path to the real system. This validates the fundamental security principle that the question is not *if* you will be breached, but *what happens when you are*. Proper segmentation transforms a potential compromise into an intelligence-gathering opportunity.

6.6 Threat Intelligence Is Generated, Not Purchased

Every data point in this report was generated by real adversarial activity against real infrastructure, processed through a self-hosted SIEM stack, and correlated using open-source tooling. The cross-infrastructure correlation finding — identifying the same Polish-registered /29 block attacking both a cloud VPS and a residential honeypot simultaneously — is the type of threat intelligence that security teams pay commercial vendors to provide. It was produced here using a Raspberry Pi 4, a cheap VPS, Cowrie, Wazuh, and eight days of patience. The tools to understand the threat landscape are accessible to anyone willing to build the infrastructure. The barrier is not cost — it is the decision to look.

7. Endpoint Overview — Wazuh SIEM

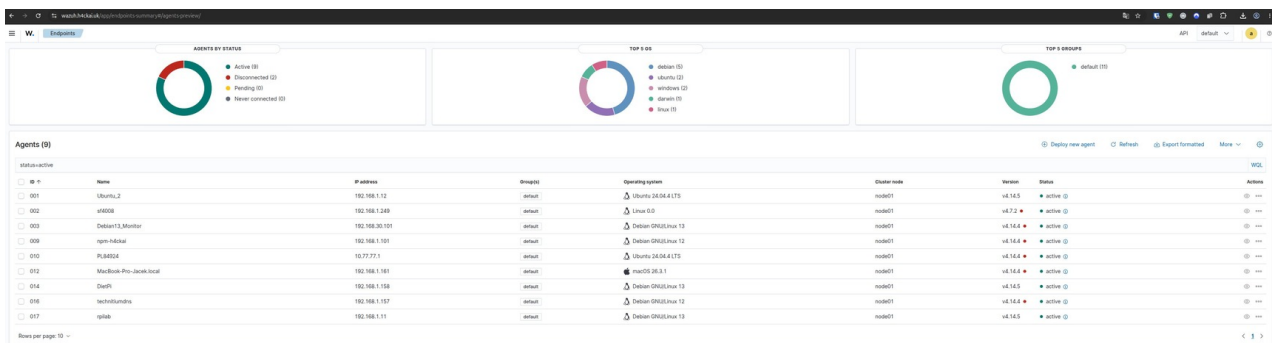


Fig 6. Wazuh Endpoints dashboard — 9 active agents across 5 operating systems (Debian, Ubuntu, Windows, Darwin, Linux)

The Wazuh SIEM instance monitors a heterogeneous fleet of 9 active endpoints across 5 operating systems — spanning IoT embedded Linux (Enigma2), Debian/Ubuntu servers, macOS (Apple M5), and Windows — all forwarding telemetry to the centralised manager via encrypted WireGuard tunnels or direct LAN connections. The rpilab (017) and PL84924 (010) agents are the two primary sensor nodes documented in this report.